

Getting to the Core

Inflation Risks within and across Asset Classes

- 인플레이션 위험
- ▶ Which assets can protect against inflation, and at what cost?
 - ▶ Conventional wisdom: currencies, commodities, and real estate are hedges, stocks are “real” assets
 - ▶ Empirically, the price of inflation risk is ambiguous
- ▶ Fama and Schwert (1977), Fama (1981), Boudoukh and Richardson (1993), Bekaert and Wang (2010), Katz, Lustig, and Nielsen (2017)

Which Inflation?

□ 어떤 인플레이션?



CPI Inflation and its components

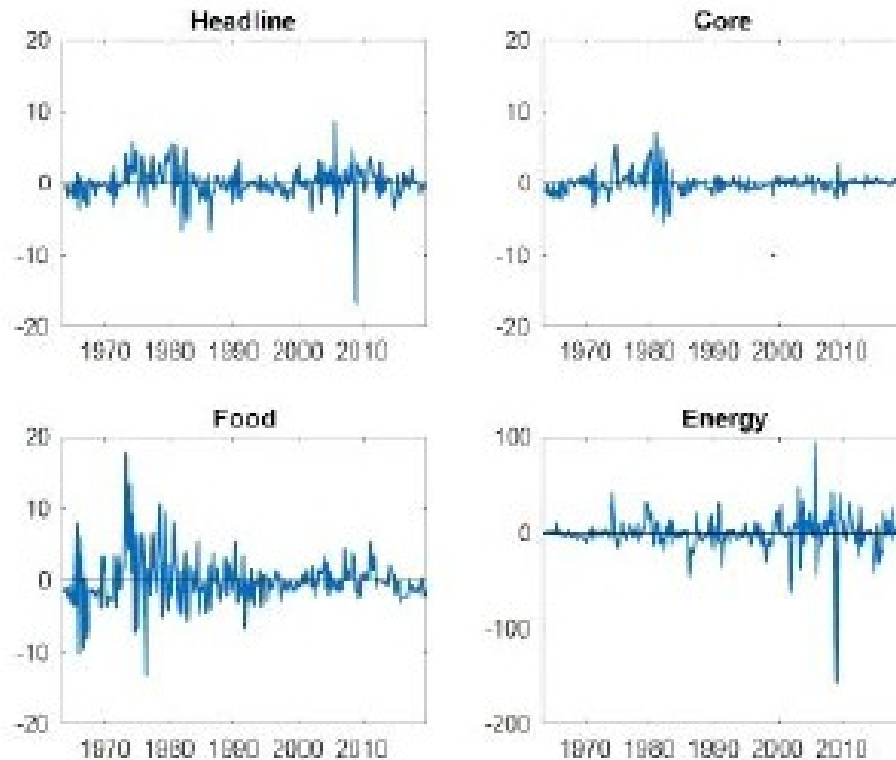
- CPI 인플레이션과 구성요소

	Headline	Core	Food	Energy
A. Summary Statistics				
Mean	3.76	3.75	3.75	4.01
Std	3.24	2.66	4.04	19.52
Persist	0.60	0.79	0.43	0.04
B. Contribution to Headline				
	1.00	0.71	0.20	0.09

시간에 따른 인플레이션 쇼크

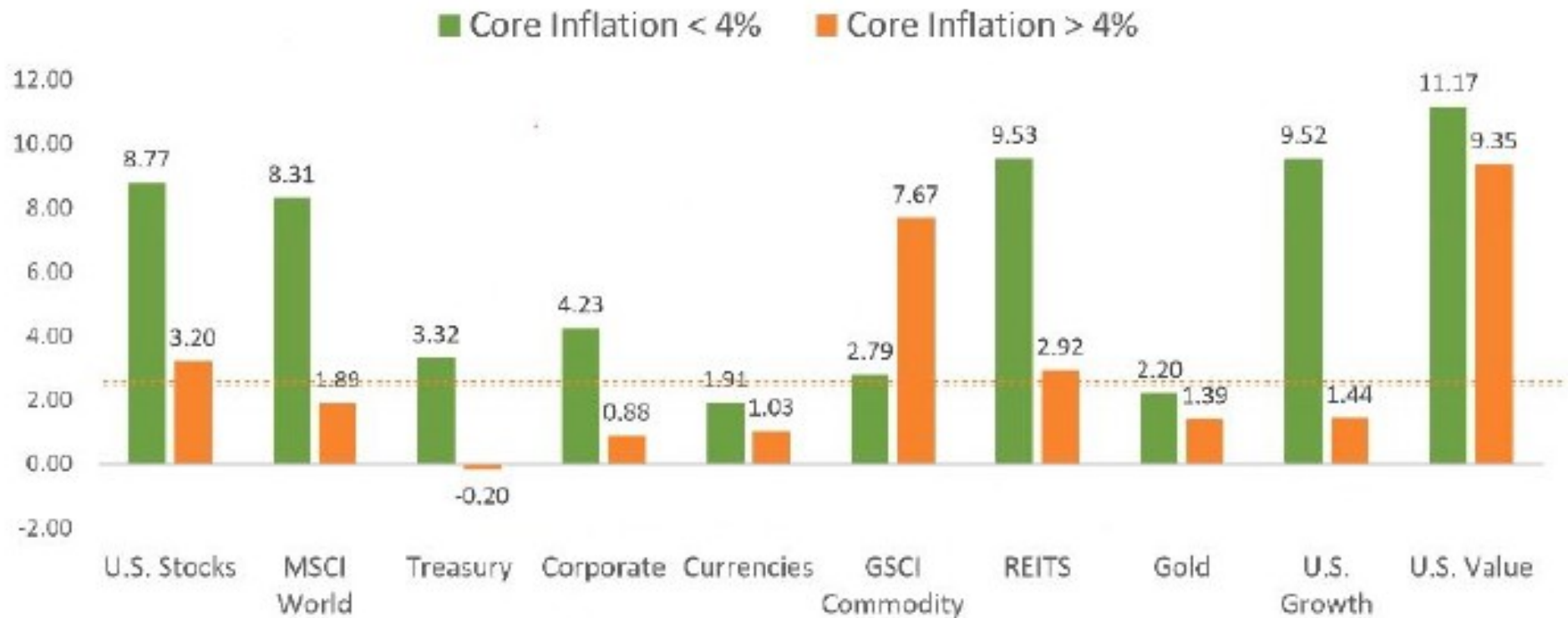
□ 시간에 따른 인플레이션 쇼크

- ▶ VAR, $Y_t = c + AY_{t-1} + u_t$, u_t as shocks
- ▶ Y_t includes headline inflation and its components, p/d ratio, risk-free rate, and output gap



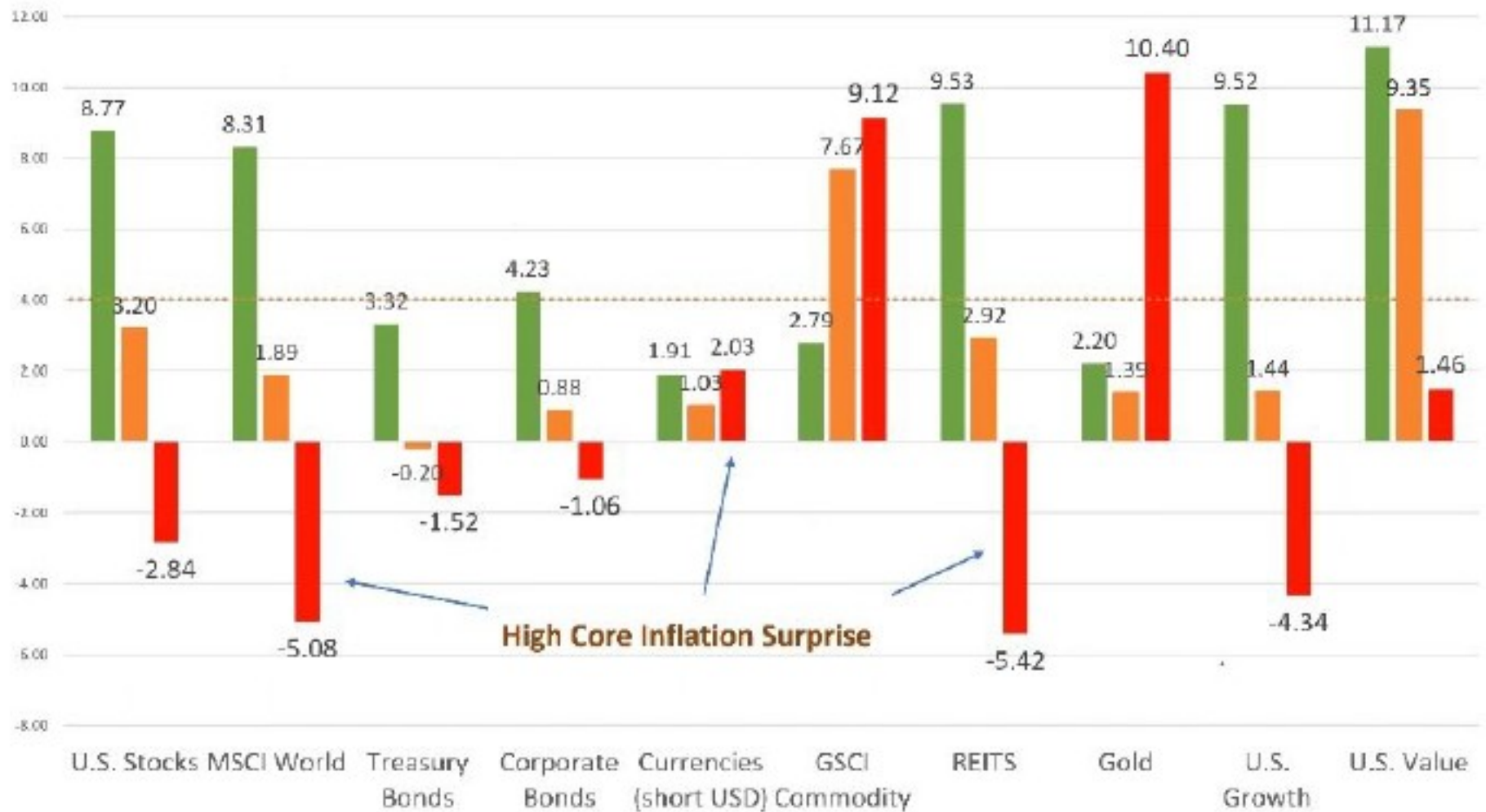
Returns when Core Inflation is Low or High

□ 근원 인플레이션



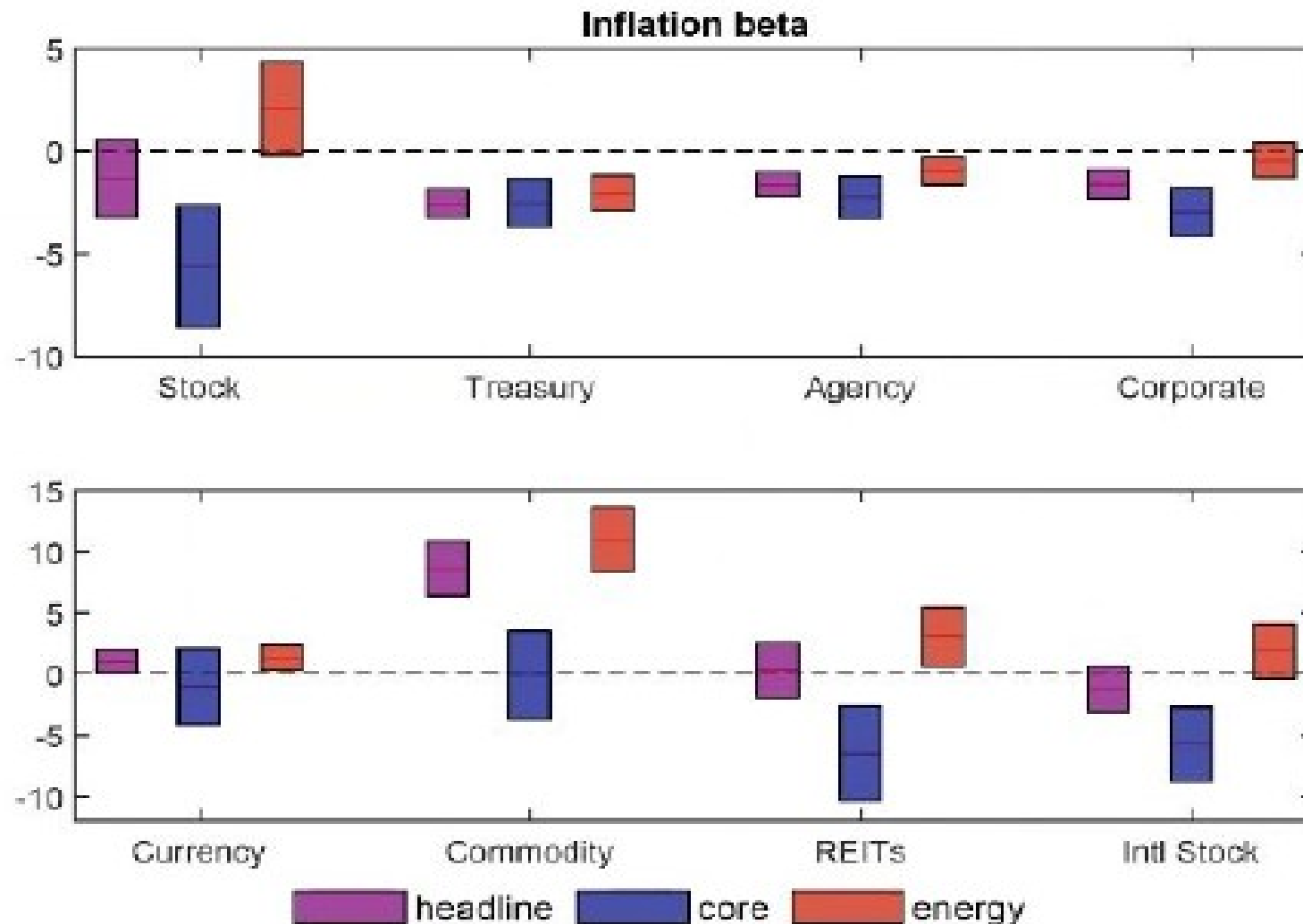
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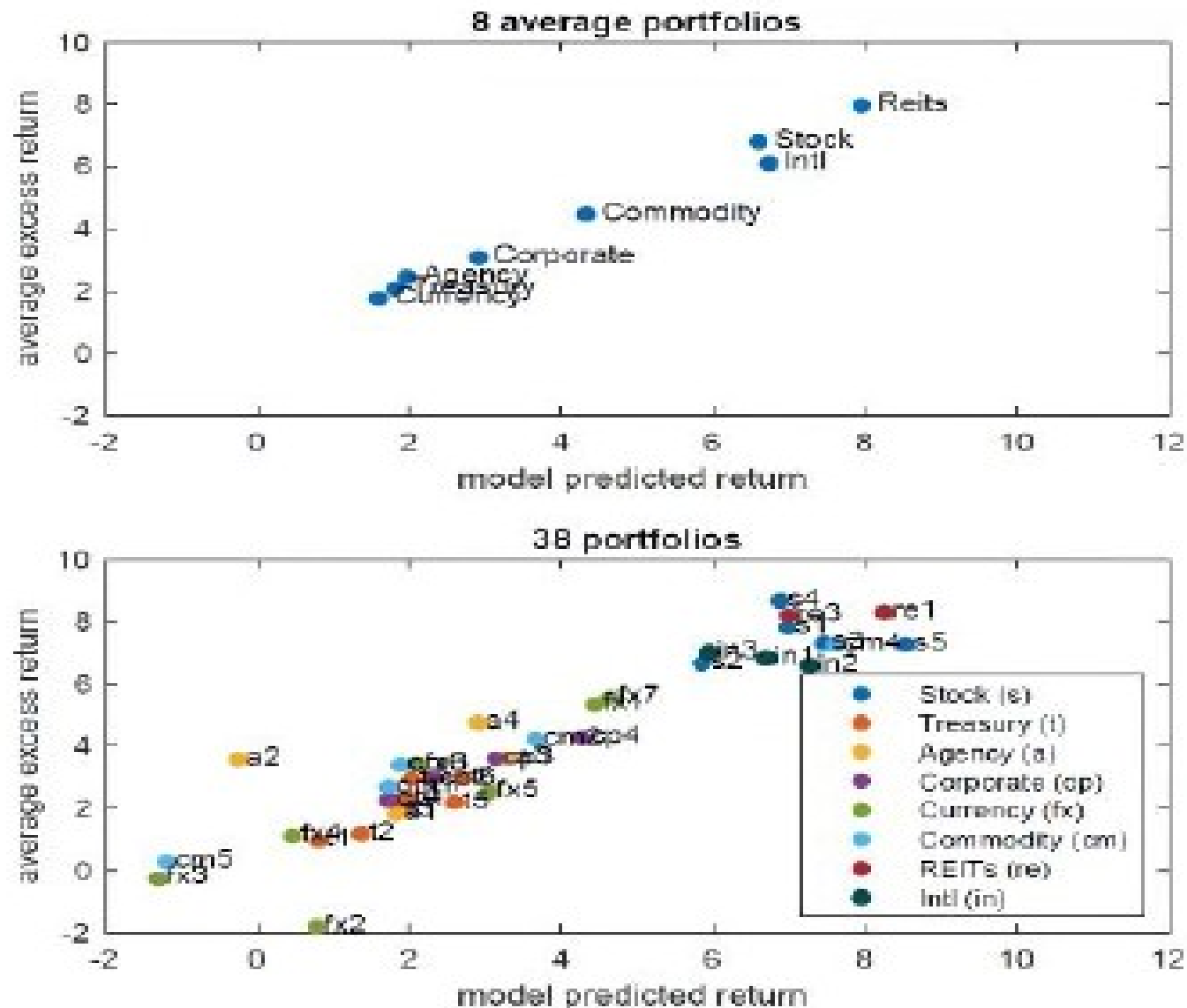
Asset Class Betas with Inflation Shocks

□ 인플레이션 베타



How Costly to Hedge? Beta vs Average Return

□ 인플레이션 헤지 비용



Aren't Stocks "Real" Assets?

□ 주식은 "실질" 자산인가?

▶ Fama (1981) proxy effect hypothesis: stock return-inflation relation due to inflation proxying for real variables

	headline	t-stat	R^2	core	t-stat	energy	t-stat	R^2
				1 quarter				
GDP	-0.14	(-1.21)	0.02	-0.21	(-1.88)	0.00	(-0.23)	0.03
Cons	-0.22	(-2.42)	0.08	-0.22	(-2.32)	-0.01	(-0.86)	0.07
Div	-0.27	(-1.15)	0.02	-0.67	(-4.27)	0.04	(0.96)	0.06
				1 year				
GDP	-0.75	(-2.34)	0.08	-0.70	(-2.24)	-0.05	(-1.05)	0.07
Cons	-0.66	(-2.24)	0.09	-0.46	(-1.81)	-0.05	(-1.12)	0.05
Div	-1.26	(-1.12)	0.03	-2.93	(-5.78)	0.18	(0.95)	0.11

▶ Core inflation negatively predicts future GDP, consumption, and dividends, especially at 1-year horizon

Cash-flow vs Discount-rate News

□ 현금흐름 대 할인율 뉴스

▶ Return news can be decomposed into CF and DR news

$$(E_{t+1} - E_t)r_{t+1} = \underbrace{(E_{t+1} - E_t) \sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j}}_{N_{CF}} - \underbrace{(E_{t+1} - E_t) \sum_{j=1}^{\infty} \rho^j r_{t+1+j}}_{N_{DR}}$$

	Cash Flow News				Discount Rate News			
	Core β	t -stat	Energy β	t -stat	Core β	t -stat	Energy β	t -stat
Mkt	-2.14	(-4.12)	-0.01	(-0.23)	4.23	(3.47)	-0.19	(-2.05)
Gr	-4.96	(-5.58)	-0.11	(-1.60)	2.57	(2.57)	-0.24	(-3.14)
BM2	-2.44	(-2.83)	0.00	(-0.03)	3.07	(3.55)	-0.10	(-1.55)
BM3	-2.28	(-2.76)	-0.03	(-0.47)	2.73	(3.37)	-0.14	(-2.26)
BM4	0.71	(0.80)	0.12	(1.80)	6.27	(4.33)	-0.12	(-1.07)
VI	1.27	(1.17)	0.08	(0.92)	7.17	(4.35)	-0.14	(-1.09)

▶ For the stock market portfolio, negative core betas come from both CF and DR news, positive energy betas mainly come from DR news

Stock-Bond Correlation over Time

□ 시간에 따른 주식-채권 상관관계의 변화

- ▶ Stock-bond correlation turned negative after 1999 (Song, 2016; Campbell et al, 2017)

	A. Headline		B. Core and Energy			
	headline	t-stat	core	t-stat	energy	t-stat
1963-1999						
Stock	-5.42	(-4.20)	-5.19	(-3.26)	-0.24	(-1.01)
Treasury	-2.88	(-5.52)	-2.77	(-4.31)	-0.20	(-2.03)
2000-2019						
Stock	2.96*	(2.22)	-6.30	(-1.18)	0.35*	(2.63)
Treasury	-2.23	(-4.73)	-0.29	(-0.15)	-0.22	(-4.65)

Note: * indicates a significant change across the two subsamples.

- ▶ Inflation and asset returns
 - ▶ First subsample: negative for stocks and bonds
 - ▶ Second subsample: positive for stock, negative for bonds
 - Driven by energy (switched signs, increased contribution)

Other “Real” Asset Classes?

□ 기타 “실질” 자산 클래스

- Foreign exchange
- Commodity Futures
- Real Estate

Currency Carry Trade Portfolios

□ 통화 캐리 트레이드 포트폴리오

		A. Headline			B. Core and energy		
	Mean	Headline β	t -stat	core β	t -stat	energy β	t -stat
			Currency				
Dcarry	5.34	-0.98	(-1.52)	-4.17	(-2.08)	0.00	(-0.04)
Carry-1	-1.81	0.33	(0.57)	-0.52	(-0.28)	0.06	(0.95)
Carry-2	-0.25	1.60	(2.99)	1.72	(1.03)	0.14	(2.55)
Carry-3	1.12	1.02	(1.92)	-0.04	(-0.02)	0.11	(2.02)
Carry-4	2.53	0.45	(0.74)	-2.50	(-1.34)	0.10	(1.60)
Carry-5	3.43	1.44	(2.28)	-1.28	(-0.65)	0.19	(2.94)
Carry-6	5.56	1.38	(1.87)	-3.62	(-1.60)	0.20	(2.72)

US Dollar

- USD
- USD **appreciates** when U.S. core inflation rises
 - after controlling for the role of energy
- Monetary channel - Fed tightening (as they are now)
- Real Rate channel - domestic goods more valuable as RGDP falls

Commodity Futures

□ 상품 선물

- Commodities: offsetting effects of price level and real growth

		A. Headline		B. Core and Energy			
	Mean	Headline β	t -stat	Core β	t -stat	Energy β	t -stat
Agriculture	0.28	4.20	(3.28)	2.06	(0.96)	0.26	(1.66)
Energy	7.26	16.51	(7.05)	-0.76	(-0.11)	1.78	(7.54)
Ind metals	4.23	4.73	(2.98)	-1.07	(-0.39)	0.66	(3.66)
Livestock	2.70	1.24	(1.24)	-1.09	(-0.66)	0.15	(1.22)
Pre metals	3.41	3.28	(2.65)	-0.22	(-0.11)	0.43	(2.96)
Gold	1.98	2.14	(1.97)	1.74	(0.91)	0.24	(1.92)
Silver	3.52	4.95	(2.63)	-0.09	(-0.03)	0.68	(3.06)
Platinum	4.36	3.40	(2.29)	7.51	(1.63)	0.26	(1.69)

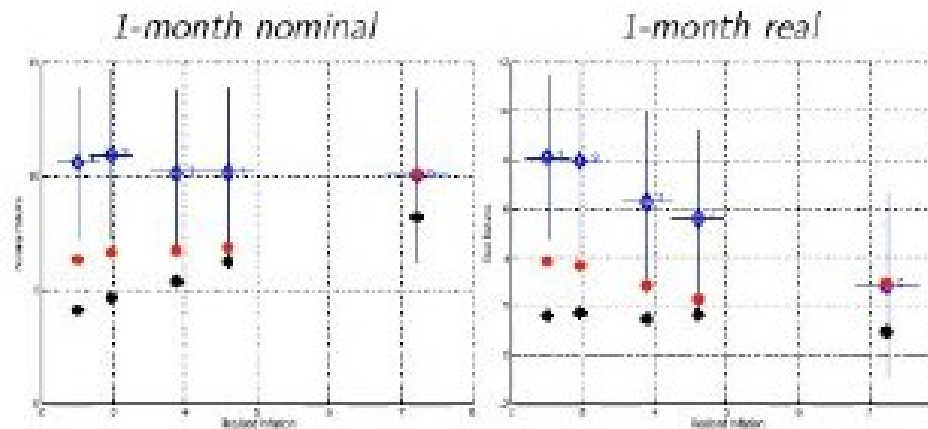
결론

- 인플레이션 리스크의 본질에 대해 새로운 시각 제공: 근원 및 에너지
 - ▶ 기존의 인플레이션 "헤지"는 에너지 인플레이션만 보호할 뿐, 근원 인플레이션은 보호하지 않음
 - ▶ 근원 인플레이션은 자산 클래스 내외에서 일관되게 추정되는 부의 리스크 프리미엄을 가진다.
- 변화하는 주식-채권 상관관계에 대한 새로운 통찰을 제시한다.
 - ▶ Shed new light on the nature of inflation risk: core and energy
 - ▶ Conventional inflation "hedges" only protect against energy inflation, not the core inflation
 - ▶ Core inflation carries a negative risk premium, consistently estimated within and across asset classes
 - ▶ New insights into the changing stock-bond correlation

Getting to the Core 해설

Incomplete Pass Through of Inflation into Nominal Returns

□ 인플레이션의 명목 수익률로의 불완전한 전가"



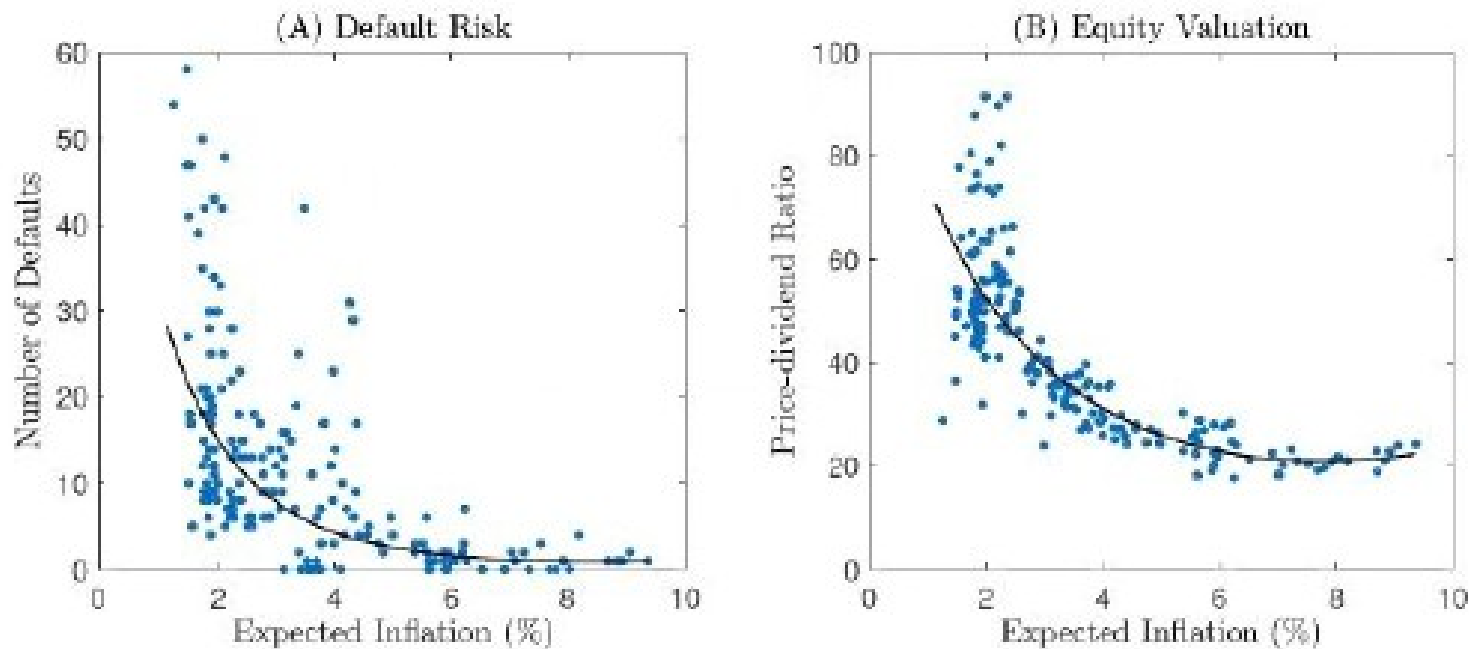
Notes: Portfolios of countries sorted by lagged inflation. We plot stock returns ('diamonds'), bond returns ('circles'), and returns on T-bills ('stars').

Source: Katz, Lustig, Nielsen (RFS, 2017)

- Incomplete pass through of local inflation into nominal returns
- Consistent w sticky discount rates or nominal cash flow extrapolation

High Expected Inflation, Default, and Valuation

□ 높은 기대 인플레이션, 디폴트 및 밸류에이션

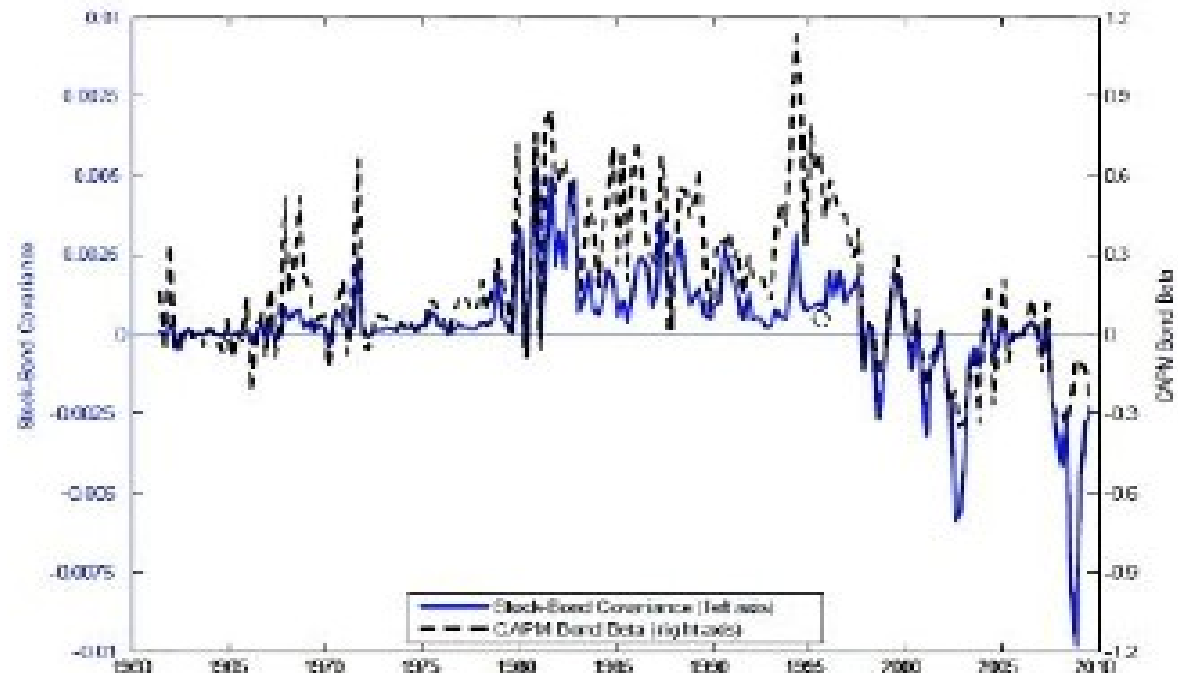


Source: Bharma, Dorion, Jeanneret, and Weber (RFS, 2022)

- High expected inflation periods of low default and low equity valuation
- Due to sticky cash flows and leverage

Changing Sign in the Stock-Bond Covariance

□ 주식-채권 공분산의 변화하는 부호



Source: Campbell, Sunderam, and Viceira (2015)

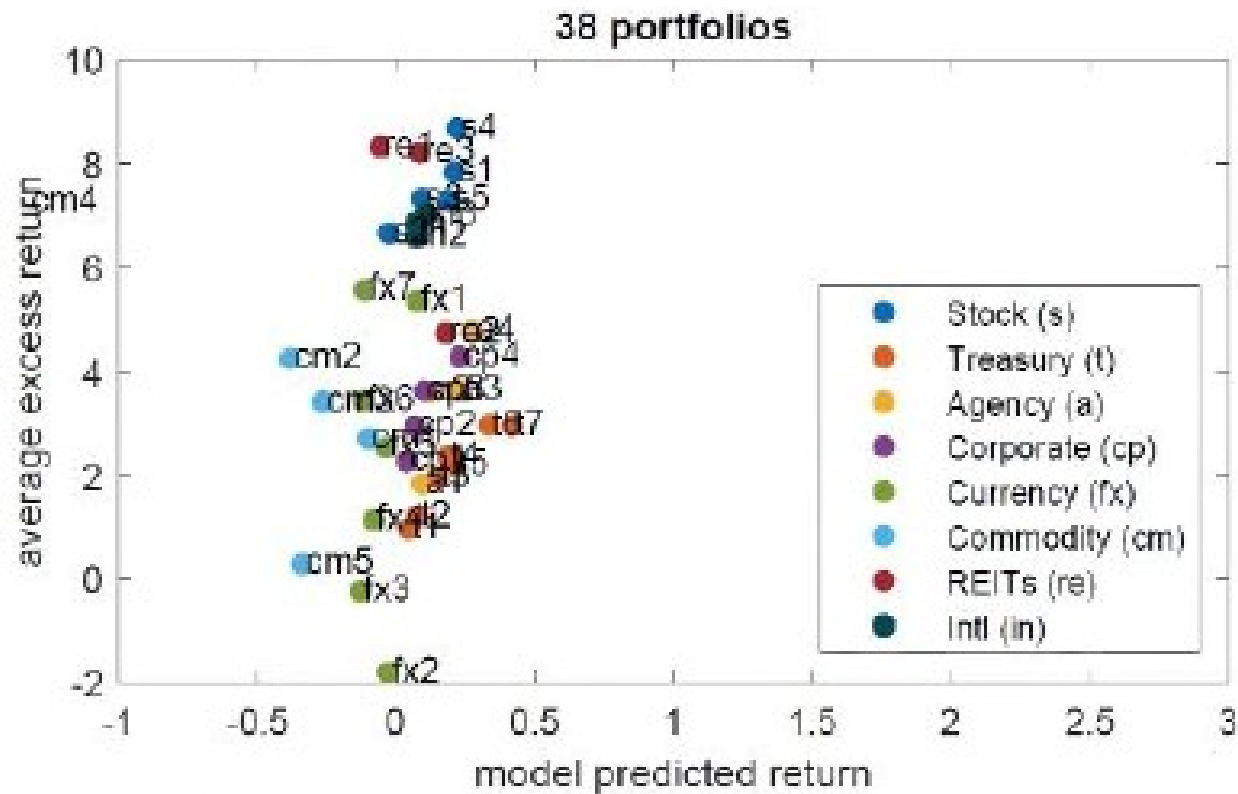
- Changing sign in the stock-bond covariance
- Stocks and bonds positively correlated in the pre 2000 period
- Coincides with change in conduct of monetary policy
- Also change in cyclical properties of inflation process

What are Inflation Hedging Properties of Real Assets?

- Boons, Duarte, de Roon, and Szymanowska (JFE, 2020)
- ▶ 인플레이션 리스크는 주식 수익률에 가격이 반영된다.
- ▶ 무조건부적 인플레이션 리스크의 가격은 작다.
- ▶ 인플레이션 리스크 프리미엄은 횡단면(XS)과 시계열(TS)에서 시간에 따라 변한다.
- ▶ 인플레이션 리스크 프리미엄은 2000년 초에 부호가 변한다.
- ▶ 시간에 따른 변동성은 리스크의 가격과 양 모두에 기인한다.
- ▶ 인플레이션은 실제 소비 성장률을 예측하기 때문에 가격에 인플레이션이 반영된다.
- ▶ 예측 가능성의 역사적 변화가 시간 변동성을 설명한다.

Headline Inflation Betas and the XS

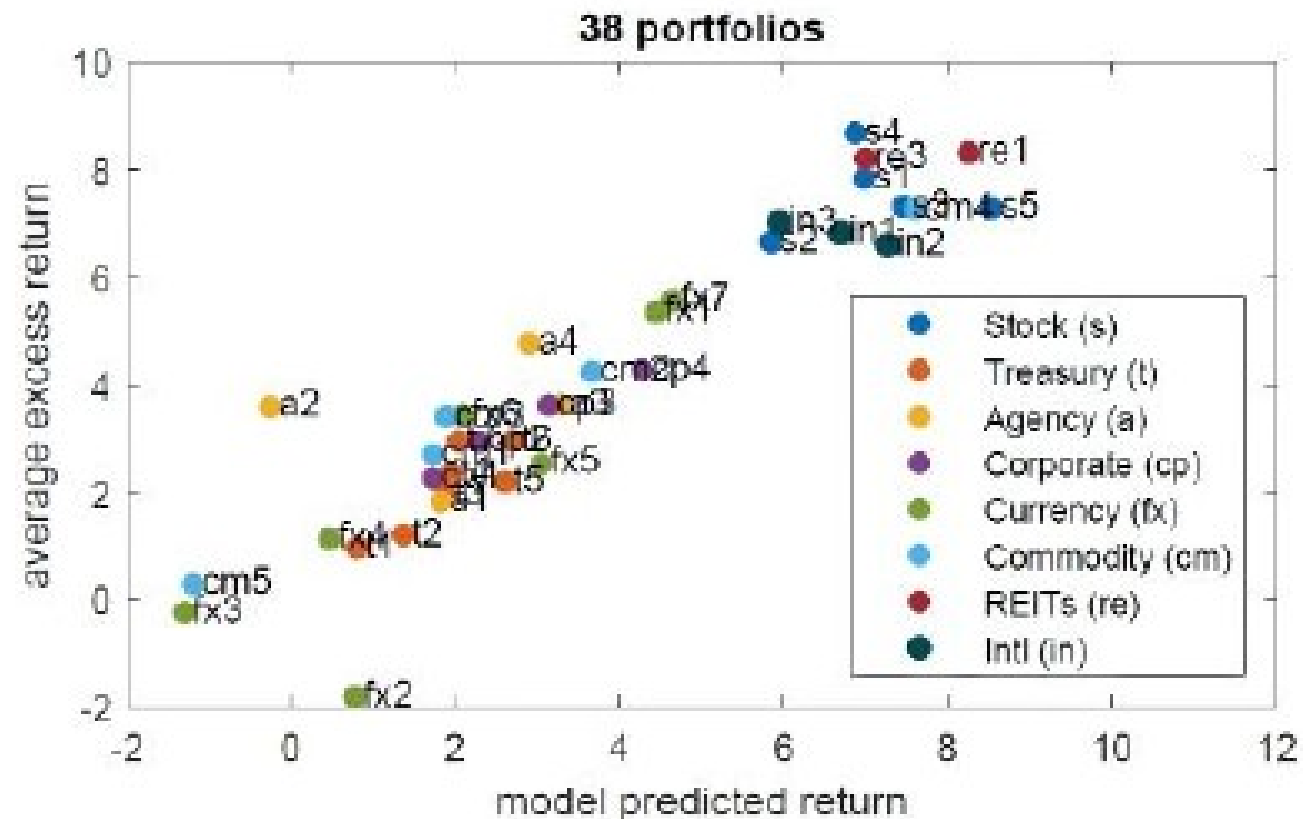
□ 헤드라인 인플레이션 베타와 횡단면 분석



■ Overall inflation betas unrelated to spread in returns

Core Inflation Betas and the XS

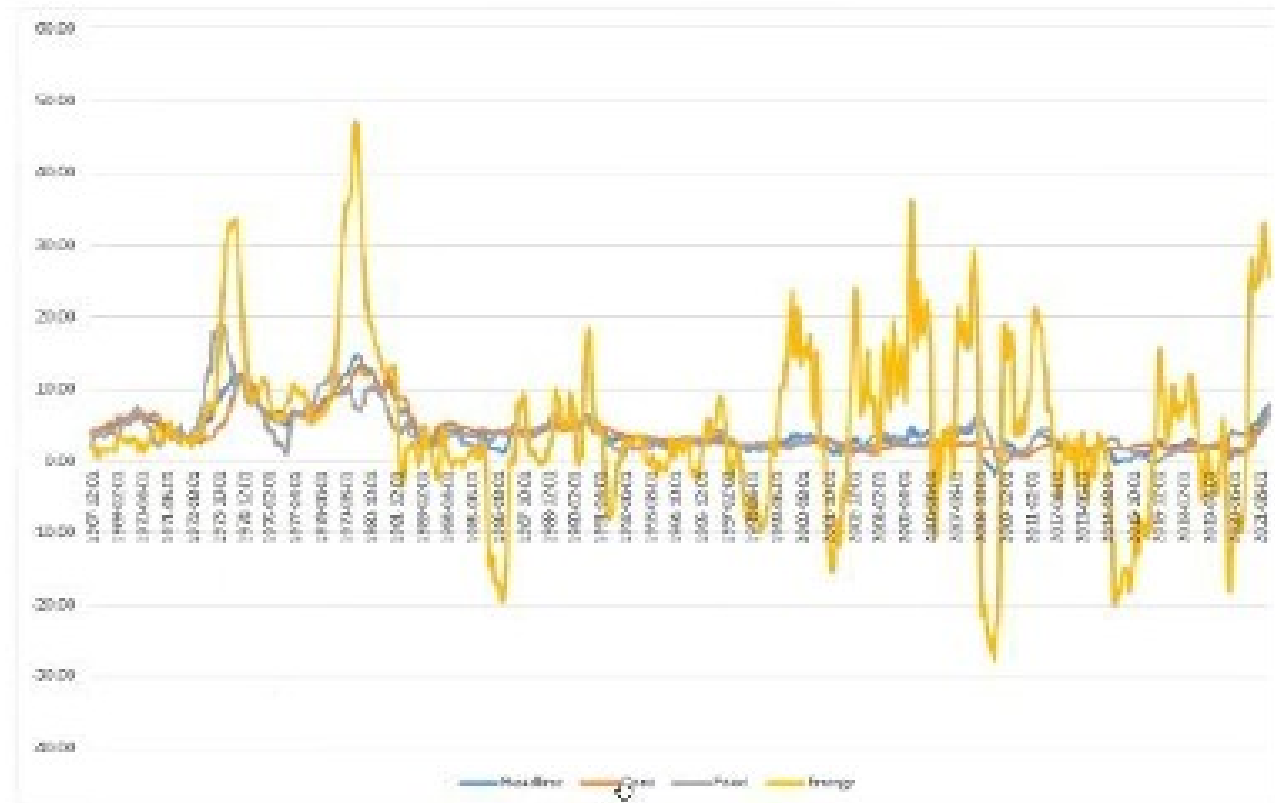
□ 근원 인플레이션 베타와 횡단면 분석



■ Core inflation betas line up with differences in average returns

Different Cyclical Properties of Inflation

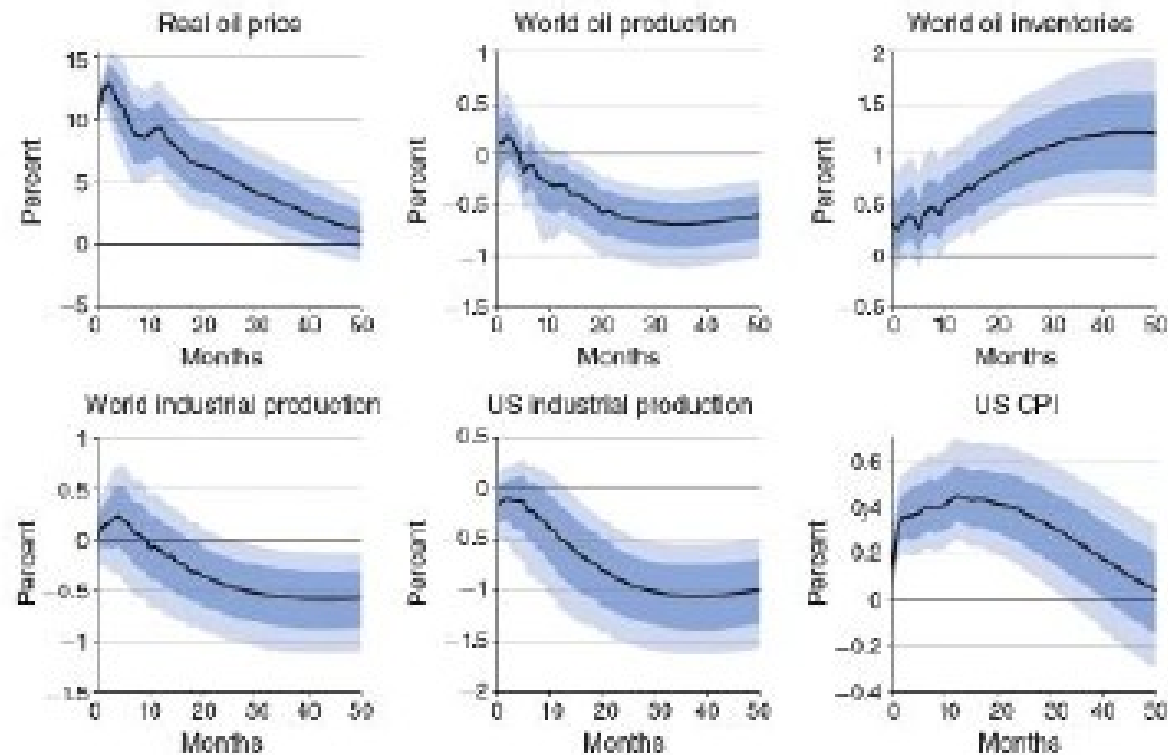
□ 인플레이션의 여러 경기순환적 특성



- Energy (and food) inflation volatile and quickly mean reverting
- Change in cyclical property of overall inflation over time
- Low interest rate environment since 2008

Pass Through of Energy Inflation into Headline Inflation

□ 에너지 인플레이션의 헤드라인 인플레이션으로의 전이"

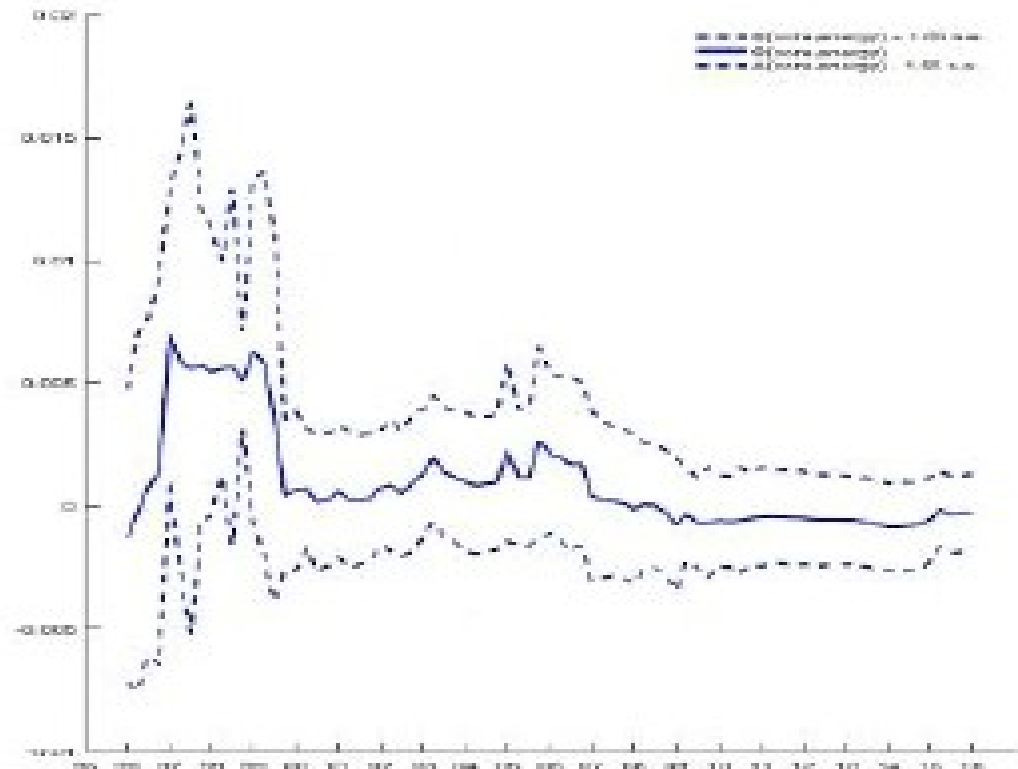


Source: Kaenzig (AER, 2021)

- Oil news shock increases US headline inflation

Pass Through of Energy Inflation into Core Inflation

□ 에너지 인플레이션의 근원 인플레이션으로의 전가



Source: Ajello, Benzoni, and Chyruk (RFS, 2020)

- Zero pass through of energy inflation into core inflation

Inflation and Monetary Policy

□ 인플레이션과 통화 정책

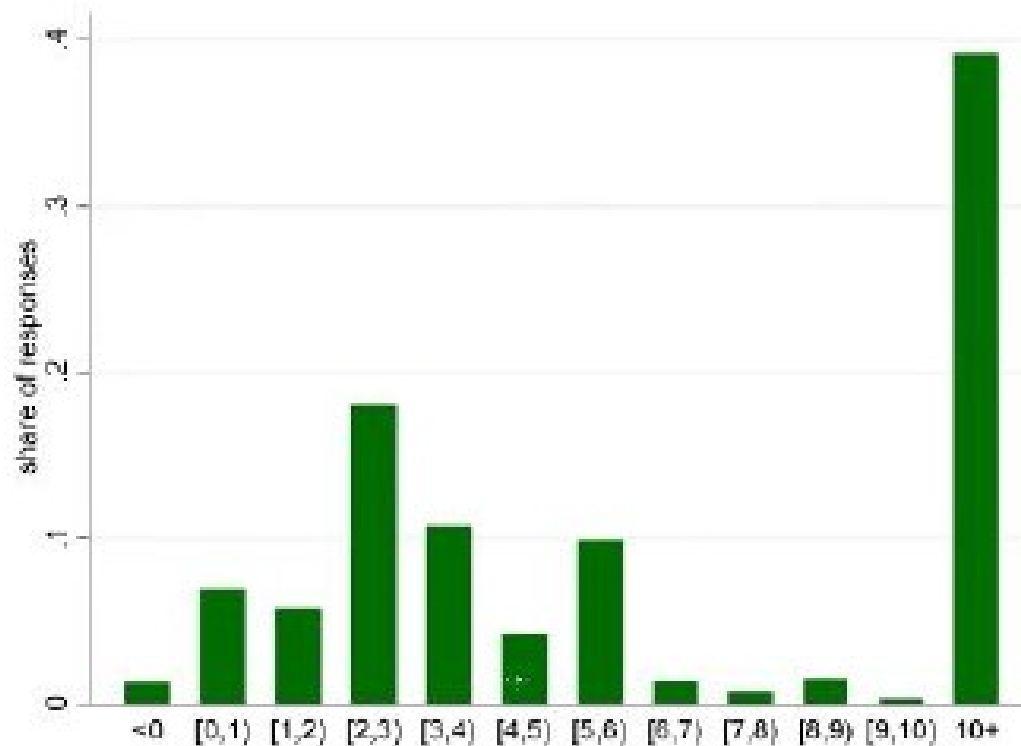
- Non-core inflation highly mean reverting
- Little predictive power for future inflationary pressure
- Monetary policy: focus on core inflation for inflationary pressures
- Higher core inflation → higher policy rates
- “Of course, if commodity and energy prices were to lead to a general expectation of a broader increase in inflation, more substantial policy rate increases would be justified. But assuming there is a generally high degree of centralbank credibility, there is no reason for such expectations to develop”

Charly Evans, President and CEO of Chicago Fed

BUT Households and Firms Lack Basic knowledge

□ 하지만 가계와 기업은 기본적인 지식이 부족하다

Inflation Target of the Fed in 2018 as perceived by 25,000 Americans

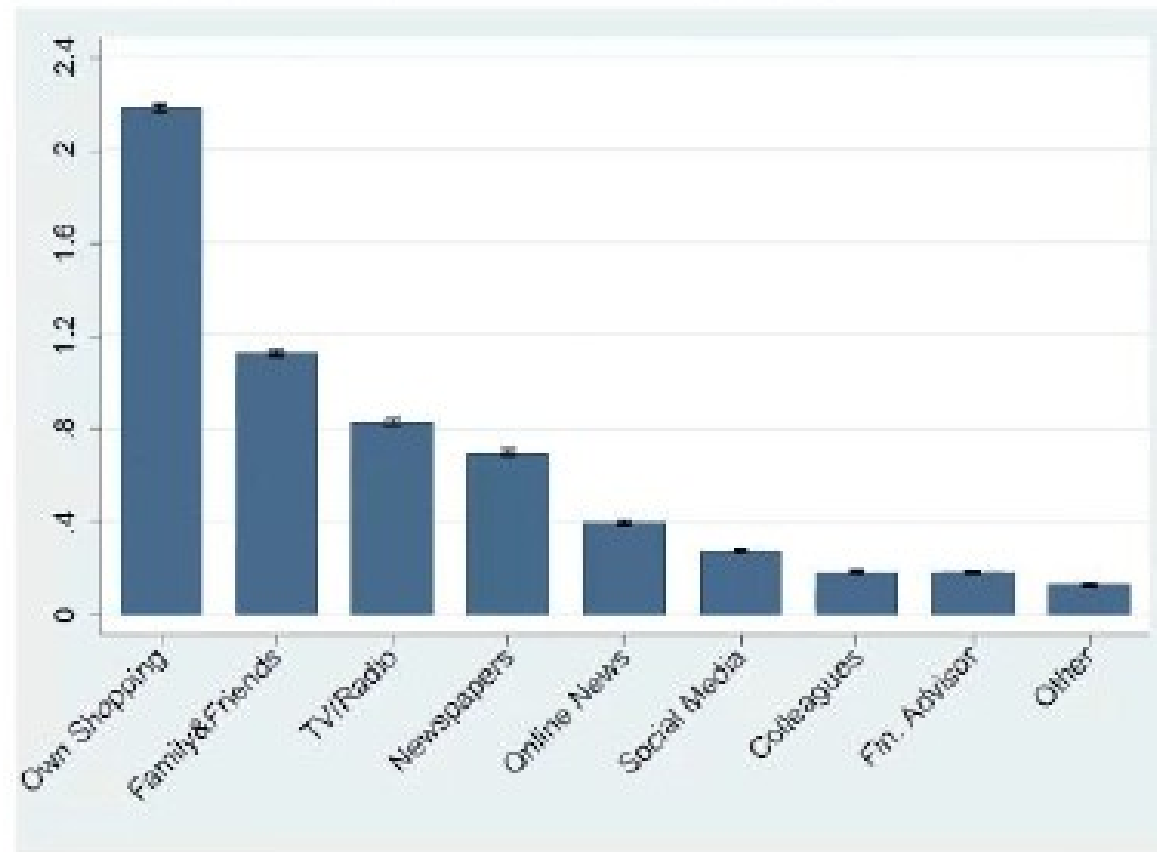


Source: Coibion, Gorodnichenko, Weber, JPE (2022):

■ Expectations react to info and determine consumption choices

BUT How Do Agents Form Expectations?

□ 하지만 경제 주체들은 기대를 어떻게 형성하는가?

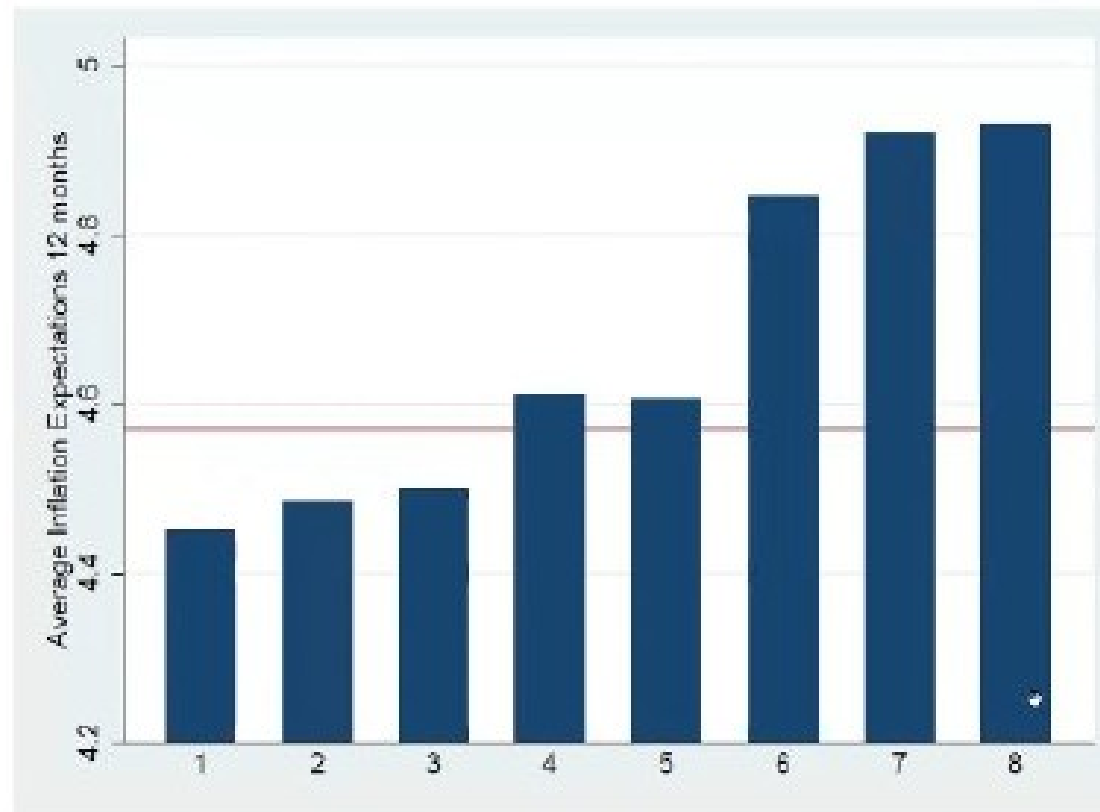


Source: D'Acunto, Malmendier, Ospina, and Weber (JPE, 2021)

■ Shopping most important source of information for future overall inflation

BUT How Do Agents Form Expectations?

□ 하지만 경제 주체들은 기대를 어떻게 형성하는가?



Source: D'Acunto, Malmendier, Ospina, and Weber (JPE, 2021)

■ Variation in shopping bundle inflation → heterogeneity in overall inflation expectations

Monetary Policy Response as Driver

□ 주요 동인으로서의 통화 정책 대응

- Core inflation betas systematically negative
 - Because of more than one for one response of monetary policy?
 - Higher core inflation triggers increase in real interest rates
 - Future consumption growth should be lower
 - Authors run event study of inflation news on FFF and stock futures
 - Stock futures have negative reaction controlling for FFF
 - But empirically, tighter monetary policy also increases risk premia
- Bernanke and Kuttner (JF, 2005)
- Negative core betas might still be reflection of policy reaction

Is Higher Inflation Always Bad?

□ 고인플레이션은 항상 나쁜가?



Source: Gourio and Ngo (2020)

- Higher inflation expectations can comove positively with higher stock valuations
- During binding ZLB, higher inflation does not trigger increase in policy rates
- Higher inflation expectations hence lower real interest rates (Fisher equation)
- Lower real interest rates stimulate aggregate demand (Euler equation)

Subsample Analysis

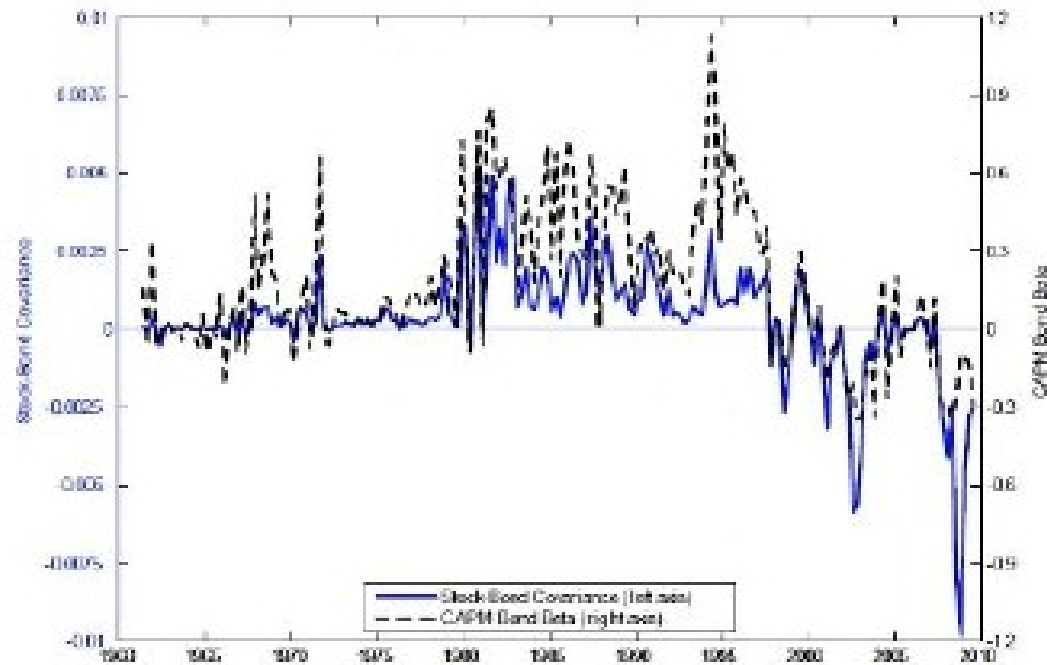
□ 부분샘플 분석

	A. Headline		B. Core and Energy			
	headline β	t -stat	core β	t -stat	energy β	t -stat
1963-1999						
Stock	-5.42	(-4.20)	-5.19	(-3.26)	-0.24	(-1.01)
Treasury	-2.98	(-5.52)	-2.77	(-4.31)	-0.20	(-2.03)
Agency	-2.49	(-4.31)	-2.50	(-3.56)	-0.12	(-1.30)
Corporate	-3.23	(-5.76)	-3.05	(-4.44)	-0.22	(-2.22)
Currency	0.12	(0.08)	0.26	(0.09)	0.06	(0.33)
Commodity	4.55	(2.80)	0.42	(0.21)	0.65	(2.24)
REIT	-5.79	(-3.76)	-5.91	(-3.29)	-0.37	(-1.43)
Intl Stock	-5.92	(-4.60)	-5.12	(-3.24)	-0.46	(-1.97)
2000-2019						
Stock	2.96	(2.22)	-0.30	(-1.18)	0.35	(2.03)
Treasury	-2.23	(-4.73)	-0.29	(-0.15)	-0.22	(-4.65)
Agency	-1.05	(-4.38)	-0.27	(-0.27)	-0.10	(-4.11)
Corporate	-0.20	(-0.49)	-0.66	(-0.40)	-0.01	(-0.22)
Currency	1.30	(2.67)	-1.87	(-0.96)	0.15	(3.10)
Commodity	12.44	(8.35)	-0.38	(-0.06)	1.24	(8.44)
REIT	3.47	(2.22)	-10.18	(-1.65)	0.44	(2.88)
Intl Stock	3.28	(2.47)	-6.62	(-1.25)	0.39	(2.95)

- Headline betas change sign over time
- Energy betas also flip signs
- Core betas consistently negative

Changing Sign in the Stock-Bond Covariance

□ 주식-채권 공분산의 변화하는 부호



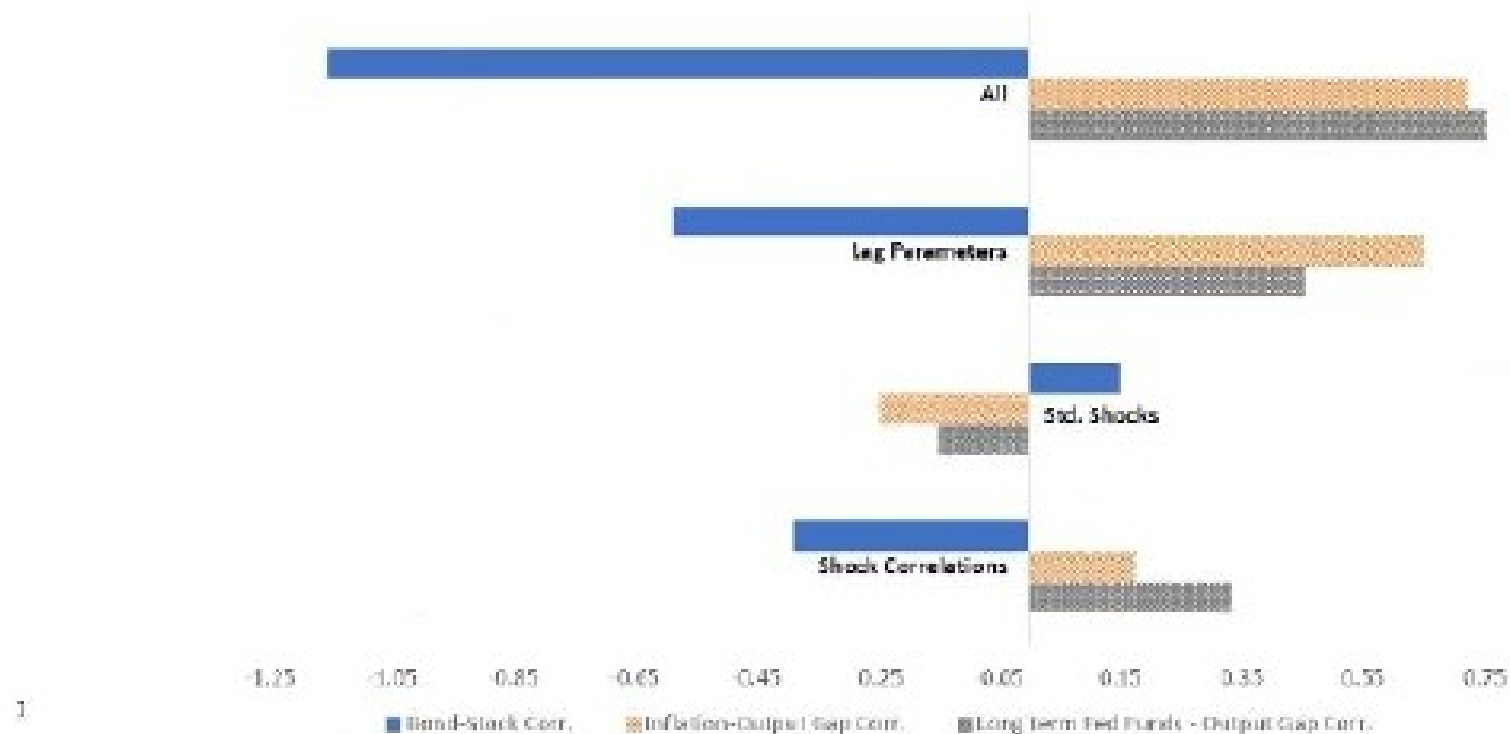
¹ Source: Campbell, Sunderam, and Viceira (2015)

- Changing sign in the stock-bond covariance
- Stocks and bonds positively correlated in the pre 2000 period
- Coincides with change in conduct of monetary policy
- Also change in cyclical properties of inflation process

Monetary Policy as Driver

□ 주요 동인으로서 통화정책

Figure 6: Counterfactuals



Source: Campbell, Pflueger, and Viceira (JPE, 2020)

- Changing parameters in policy rule biggest driver of change in stock-bond correlation

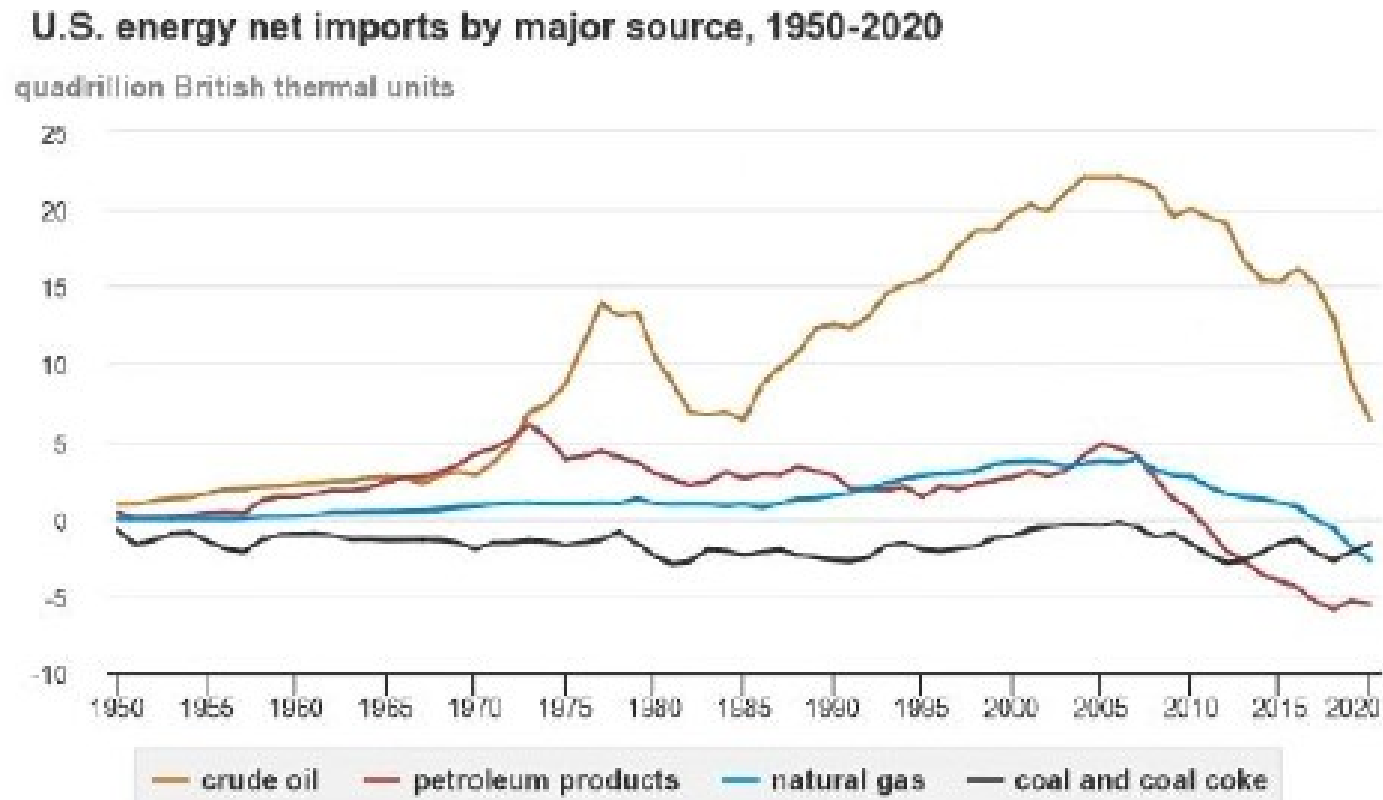
Drivers of Inflation

□ 인플레이션의 주요 동인

- Energy inflation often due to supply shocks such as wars, etc
- Hence, energy inflation bad for US?
- Depends on whether US is net exporter or importer of energy

Net Energy Imports

□ 순 에너지 수입



Source: U.S. Energy Information Administration, *Monthly Energy Review*, Table 1.4 c, April 2021, preliminary data for 2020

- Large decrease in net imports of crude oil since 2000

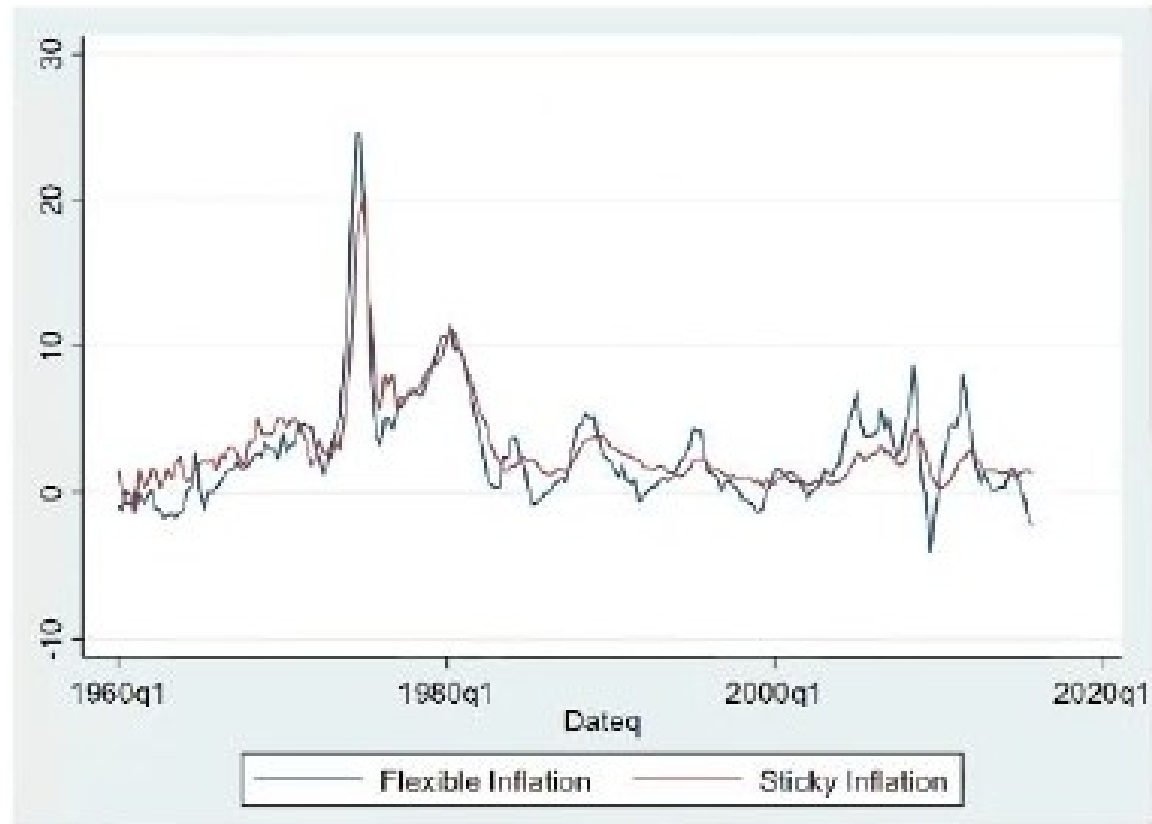
Core versus Energy Inflation

□ 근원 대 에너지 인플레이션

- Authors conjecture price stickiness as fundamental distinguishing force
- Use frequencies of price adjustments from data underlying PPI
Gorodnichenko and Weber (AER, 2016)
- Merge frequencies with sectoral PPI inflation at 6-digit NAICS level
- Calculate sticky and flexible inflation based on median frequency

Sticky and Flexible Sector Inflation

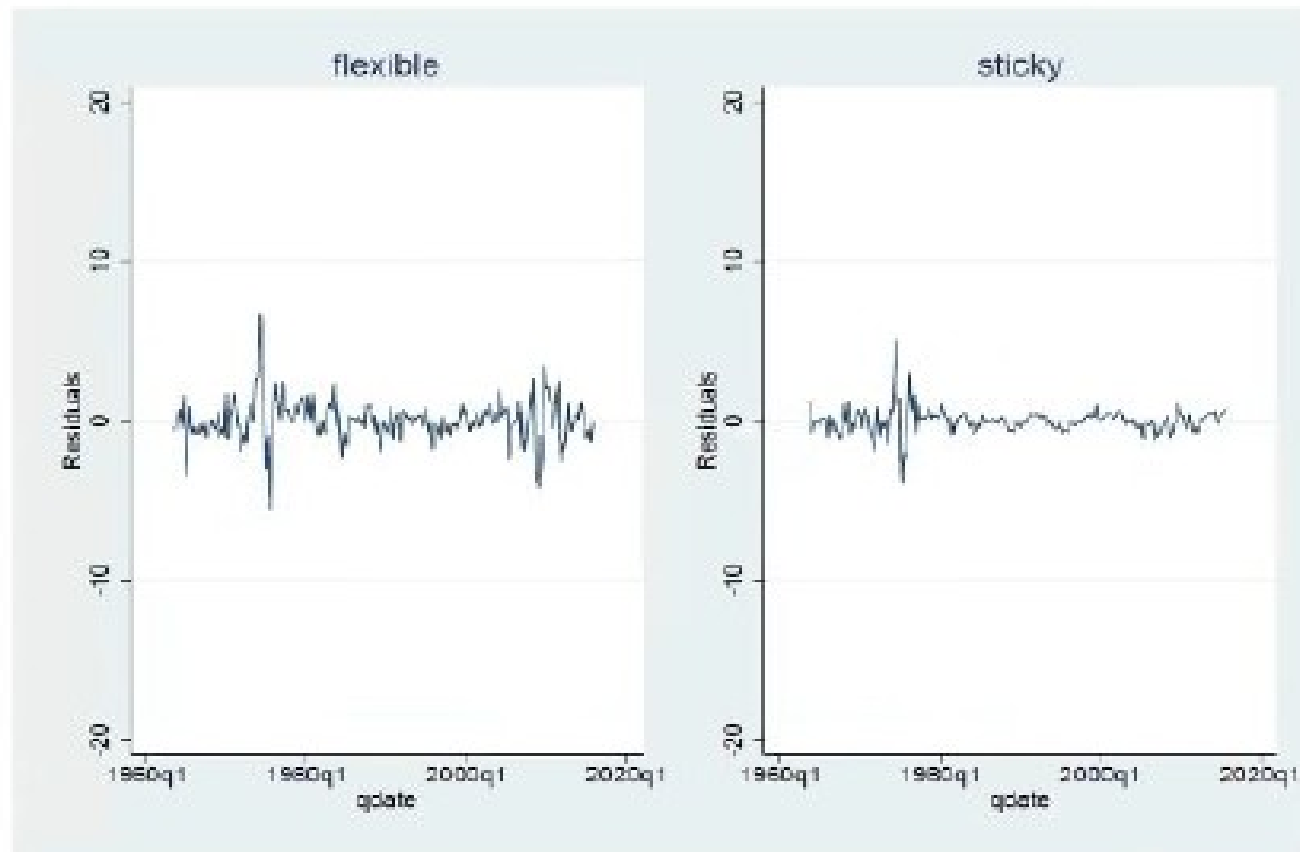
□ 경직적 및 유연한 섹터 인플레이션



■ Flexible inflation more volatile consistent with energy inflation

Sticky and Flexible Sector Inflation

□ 경직적 및 유연한 섹터 인플레이션



- Flexible inflation series highly correlated with energy inflation (64%)
- Sticky inflation series highly correlated with core inflation (86%)

Model: Price of Energy Inflation

□ 에너지 인플레이션의 비용

- Price of risk for energy inflation ambiguous
 - Affected by both demand and supply shocks
- pre-1999: energy beta of stocks negative → supply shocks important
- Demand channel driving force in first decades of the 21st century
- Empirically aim to disentangle
- More generally, what are reduced form shocks in VAR capturing

Other issues

□ 기타 이슈

- The inflation innovation are generated regressors
- Alternative state variables in VAR
- Expanding window
- Simple differences
- Multivariate regressions also including other factors